

STANDARD OPERATING PROCEDURE

SOP-02: Infection Control & Bioburden Management

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Department: Clinical Operations & Safety
Approved By: Robert E. Beaty, Founder

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Modality: Mobile Diagnostics

1.0 Purpose & Scope

1.1 Purpose: To establish strict, evidence-based infection control protocols that eliminate the risk of cross-contamination between patients during mobile pulmonary function testing. This procedure ensures complete adherence to Centers for Disease Control and Prevention (CDC) ambulatory care guidelines and Occupational Safety and Health Administration (OSHA) bloodborne pathogen and respiratory protection standards.

1.2 Scope: This protocol applies to all diagnostic hardware, disposable consumables, and clinical environments utilized by DirectCare Pulmonary Diagnostics LLC during contracted testing blocks at partner facilities.

2.0 Primary Bioburden Barrier (Filters)

Because the PulmOne MiniBox+ captures direct forced expiratory samples, safeguarding the internal pneumotachograph and environmental air is paramount.

- **Single-Use Mandate:** DirectCare operates on a strict 100% disposable patient-interface model. No reusable mouthpieces or breathing tubes shall be utilized under any circumstances.
- **Filtration Efficiency:** Testing must exclusively utilize FDA-cleared, 0.025-micron inline viral and bacterial barrier filters. These specific filters are clinically validated to provide greater than **99.999% filtration efficiency** against both bacterial and viral pathogen transmission, entirely insulating the device interior from patient bioburden.
- **Nose Clips:** Single-use, disposable medical-grade foam nose clips are to be used for every patient and discarded immediately following the test.

3.0 Between-Patient Disinfection Workflow

To maintain the integrity of the host clinic's exam room and ensure absolute safety for the subsequent patient, the following surface disinfection protocol must be executed between every diagnostic encounter:

1. **Hand Hygiene:** The supervising Registered Respiratory Therapist (RRT) must perform hand hygiene (alcohol-based hand rub or soap and water) and don fresh, non-sterile nitrile examination gloves before initiating the breakdown of the testing interface.
2. **Filter Removal:** The used viral/bacterial filter and nose clip are to be carefully detached from the MiniBox+ sensor arm, minimizing agitation of the filter media.
3. **Surface Sanitization:** Using EPA-registered, hospital-grade germicidal wipes (e.g., CaviWipes or Super Sani-Cloth), the RRT will thoroughly wipe down all high-touch surfaces, including:
 - The exterior chassis of the PulmOne MiniBox+.
 - The flexible sensor arm and external sensor housing.
 - The mobile diagnostic cart surface and handles.
 - The patient seating area (if utilizing a DirectCare-provided stool).
4. **Contact Time:** The RRT must allow the surfaces to remain visibly wet for the manufacturer-prescribed contact/kill time (typically 2 to 3 minutes) prior to seating the next scheduled patient.
5. **Doffing:** Gloves are removed and discarded, followed immediately by secondary hand hygiene.

4.0 Regulated Medical Waste (RMW) Management & Transit Liability

To strictly comply with Department of Transportation (DOT) and Environmental Protection Agency (EPA) regulations regarding the transport of hazardous materials, DirectCare personnel are strictly prohibited from transporting Regulated Medical Waste (RMW) in commercial company vehicles.

- **Host Clinic Disposal:** As codified in the master Memorandum of Understanding (MOU) signed by all partner clinics, the host facility agrees to provide on-site access to their existing, contracted biohazardous waste management receptacles (e.g., standard "Red Bags" or bio-bins).
- **Immediate Discard:** All used barrier filters, nose clips, and soiled nitrile gloves generated during the "PFT Clinic Day" will be disposed of directly into the host clinic's approved clinical waste bins inside the exam room immediately following each test.
- **Zero Transit Policy:** Under no circumstances will biological consumables be bagged, stored, or transported back to the DirectCare headquarters.

5.0 Staff Personal Protective Equipment

While the inline barrier filters prevent device and environmental contamination, the RRT remains in close proximity to patients executing forced expiratory maneuvers.

- **Standard Precautions:** Non-sterile nitrile gloves must be worn during testing execution, equipment breakdown, and surface sanitation.

- **Respiratory Protection:** During periods of elevated regional respiratory transmission (e.g., influenza, RSV, or COVID-19 surges), or if the partner clinic maintains an active masking mandate, the RRT will wear an N95 respirator or ASTM Level 3 surgical mask while testing patients.

END OF STANDARD OPERATING PROCEDURE SOP-02